

Microeconomics

Lecture 20: General Review

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Economic profit and the optimality rule

Premise. The price-taking firm takes the price P as given. Maximizing profit means choosing the quantity at which the additional revenue from one unit ($MR = P$) equals its additional cost (MC).

Economic profit

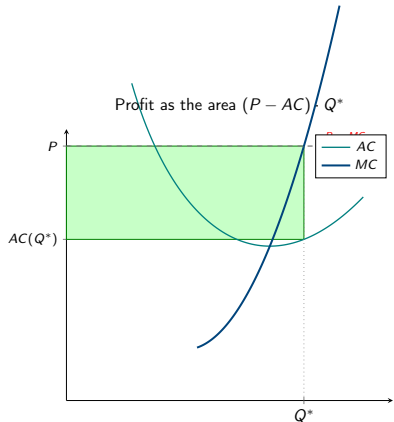
$$\pi(Q) = TR(Q) - TC(Q) = P \cdot Q - C(Q)$$

It includes (implicit) *opportunity costs*: $\pi = 0$ means a normal return, not zero earnings.

Optimality conditions (PC)

$$P = MC(Q^*) \quad \text{and} \quad MC'(Q^*) > 0$$

The second condition rules out the profit *minimum*.
Equivalently $MR = MC$ with constant $MR = P$.



Shutdown condition and the short-run supply curve

Premise. Even with $\pi < 0$ it may pay to produce in the SR: fixed costs are *sunk*. The real choice is between producing and losing less than FC , or shutting down and losing FC .

Shutdown rule (SR)

The firm produces if and only if

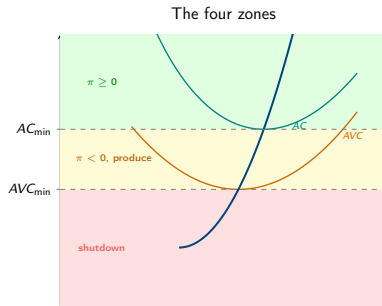
$$P \geq AVC_{\min}$$

The shutdown point is the *minimum* of AVC ; the break-even point is the *minimum* of AC . In the LR, FC are recoverable and the threshold becomes $P \geq AC_{\min}$.

Individual SR supply

$$Q^s(P) = \begin{cases} Q : MC(Q) = P, MC' > 0 & P \geq AVC_{\min} \\ 0 & P < AVC_{\min} \end{cases}$$

It is the rising portion of MC above $AVC_{\min}$, *not* the whole MC .



Horizontal aggregation and the market

Premise. The market supply curve is the *horizontal sum* of individual supply curves: at each price P , we add up the quantities each firm is willing to produce.

n identical firms

If each firm has supply $q^s(P)$ (above $AVC_{\min}$):

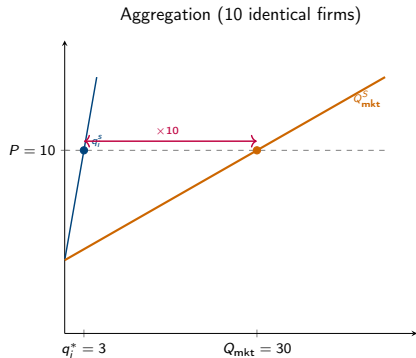
$$Q_{\text{mkt}}^s(P) = n \cdot q^s(P)$$

Same price threshold as individual supply; total quantity multiplied by n .

Heterogeneous firms

$$Q_{\text{mkt}}^s(P) = \sum_{i=1}^n q_i^s(P)$$

The aggregate curve is *piecewise*: each new threshold price activates a new set of firms. Inverse form $P = P^s(Q)$: it coincides with the aggregate MC above the shutdown thresholds.



Elasticity of supply

Definition

$$\varepsilon_S = \frac{dQ^S}{dP} \cdot \frac{P}{Q^S} = \frac{\% \Delta Q^S}{\% \Delta P} \geq 0$$

Classification

$\varepsilon_S = 0$	vertical (perf. inelastic)
$0 < \varepsilon_S < 1$	steep (inelastic)
$\varepsilon_S = 1$	unit elastic
$\varepsilon_S > 1$	flat (elastic)
$\varepsilon_S \rightarrow \infty$	horizontal

Slope $\neq$ elasticity

The slope dQ^S/dP is constant along a straight line, but ε_S changes with the point (P, Q) at which it is evaluated.

Determinants of the level of ε_S

The easier it is to expand production (spare capacity, readily available inputs, long time horizon), the more elastic supply is.

Market equilibrium and surplus

Premise. The competitive equilibrium (P^*, Q^*) is the pair that clears the market: $Q^D(P^*) = Q^S(P^*)$. It is also the point at which total welfare is maximized (*first welfare theorem*).

Consumer and producer surplus

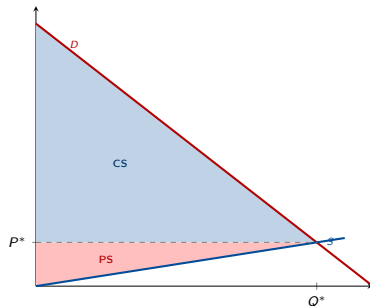
$$CS = \int_0^{Q^*} [P_D(q) - P^*] dq$$

$$PS = \int_0^{Q^*} [P^* - P_S(q)] dq$$

$$TS = CS + PS$$

Triangle formulas (linear curves)

$$CS = \frac{1}{2} Q^* (P_{\max}^D - P^*), \quad PS = \frac{1}{2} Q^* (P^* - P_{\min}^S)$$



Long-run equilibrium with free entry

Premise. With *free entry*, $\pi > 0$ attracts new firms (S shifts right, P falls); $\pi < 0$ causes exit. The process stops when profit is zero for all firms.

LR equilibrium (constant-cost industry)

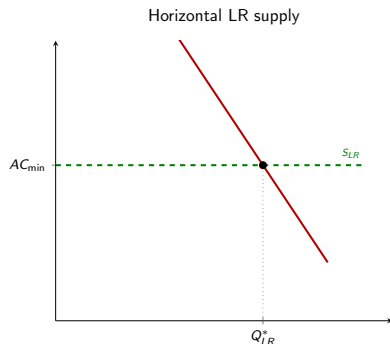
$$P_{LR}^* = AC_{\min}, \quad q^* = \arg \min AC(q), \quad \pi = 0$$

Each active firm produces *at efficient scale*.

Number of firms in the LR

$$n^* = \frac{Q^D(P_{LR}^*)}{q^*}$$

A *permanent* demand shock changes n^* , not P_{LR}^* : the LR supply curve is horizontal at $P = AC_{\min}$.



Deadweight loss: the triangle formula

Premise. Any deviation from the equilibrium quantity $\tilde{Q} \neq Q^*$ leaves efficient trades unrealized, generating a *deadweight loss* (DWL): value that goes to no one.

Formula with linear curves

$$DWL = \frac{1}{2} (Q^* - \tilde{Q}) (P_D(\tilde{Q}) - P_S(\tilde{Q}))$$

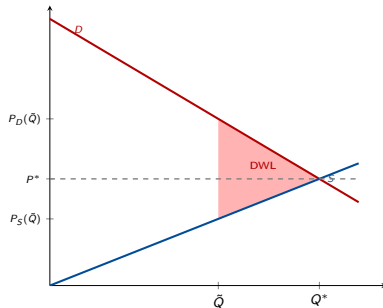
Base: quantity not traded. **Height:** difference between WTP and MC at the last unit traded.

General formula

$$DWL(\tilde{Q}) = \int_{\tilde{Q}}^{Q^*} [P_D(q) - P_S(q)] dq$$

DWL and elasticity

For a given ΔQ , the height of the triangle increases with the *steepness* of the curves (inelastic demand/supply). Taxing inelastic goods minimizes DWL for a given revenue.



Price ceiling and price floor

Price ceiling ($\bar{P} < P^*$)

Quantity traded constrained by **supply**:

$$\tilde{Q} = Q^S(\bar{P}) < Q^*.$$
 Excess demand

$Q^D(\bar{P}) - Q^S(\bar{P}) > 0$. Examples: rent control, regulated drug prices, 2022–23 energy price caps.

Price floor ($\underline{P} > P^*$)

Quantity traded constrained by **demand**:

$$\tilde{Q} = Q^D(\underline{P}) < Q^*.$$
 Excess supply

$Q^S(\underline{P}) - Q^D(\underline{P}) > 0$. Examples: minimum wage (unemployment), CAP minimum prices, minimum professional fees.

Short side and DWL

In both cases the quantity actually traded is $\tilde{Q} = \min\{Q^D, Q^S\}$ (short side). For the same $\tilde{Q}$, ceiling and floor produce the **same** DWL: only the *distribution* changes (who is rationed).

Binding or non-binding

The constraint bites only if $\bar{P} < P^*$ (ceiling) or $\underline{P} > P^*$ (floor). Otherwise the market operates at the free equilibrium, and the instrument is inert.

Specific tax: equilibrium, wedge, incidence

Premise. A specific tax t on the seller shifts supply up by t . The wedge between price paid P_d and price received P_s is exactly t .

Equilibrium with a tax

$$Q^D(P_d) = Q^S(P_d - t), \quad P_s = P_d - t$$

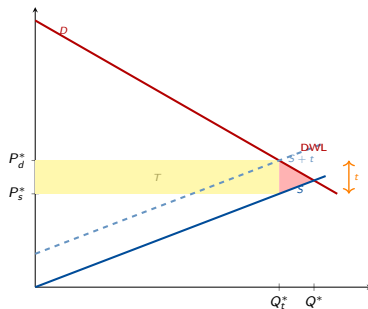
Revenue: $T = t \cdot Q_t^*$. **Wedge:** $P_d - P_s = t$.

Tax incidence

$$\text{Cons. share} = \frac{\Delta P_d}{t} = \frac{\varepsilon_S}{\varepsilon_S - \varepsilon_D}$$

$$\text{Prod. share} = \frac{|\Delta P_s|}{t} = \frac{-\varepsilon_D}{\varepsilon_S - \varepsilon_D}$$

The **less elastic** side bears the larger share.



Equivalence principle

The final equilibrium is identical whether the tax is formally levied on the seller or on the buyer. Who remits the tax is an accounting distinction.

Quadratic DWL, ad valorem tax, subsidies

DWL and the tax rate (linear curves)

$$DWL = \frac{1}{2}(Q^* - Q_t^*)t \propto t^2$$

Doubling the tax rate *quadruples* the DWL. Policy rule: *broad base, low rate*.

Ad valorem τ and subsidy s

Ad valorem: supply *rotates*, $P_d = MC/(1 - \tau)$; locally equivalent to a specific tax $t = \tau P_d^*$.

Subsidy: supply shifts *down* by s ; $P_d \downarrow$, $P_s \uparrow$, $Q \uparrow$. Cost: $s \cdot Q_s^*$; it generates DWL (trades with $WTP < MC$).

Tariffs and quotas: formal equivalence

For the same residual imports, a tariff t and a quota $\bar{Q}$ produce the *same* P , Q , CS, PS, DWL.

Difference: who captures the rent trapezoid

- **Tariff** $\rightarrow$ revenue to the government
- **Quota** $\rightarrow$ rent to license holders

The monopolist's marginal revenue

Premise. The monopolist faces the entire (downward-sloping) market demand. To sell one more unit it must lower the price *on all* units: $MR < P$.

General formula

$$MR(Q) = P(Q) + Q \cdot P'(Q) < P(Q)$$

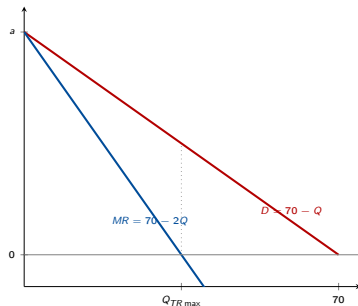
Linear demand: the twice-as-steep rule

If $P = a - bQ$: $MR = a - 2bQ$ —same intercept, twice the slope; MR crosses the Q axis halfway to D .

MR -elasticity relation

$$MR = P \left(1 - \frac{1}{|\varepsilon_D|} \right)$$

Consequence: the monopolist never produces where $|\varepsilon_D| < 1$ (it would require $MR < 0$).



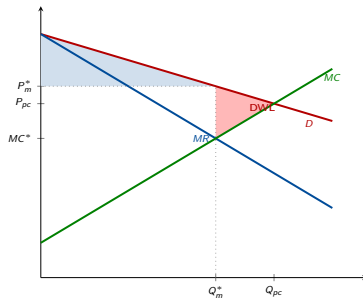
The monopolist's optimum and deadweight loss

Premise. Unlike under PC, the monopolist equates not P but *marginal revenue* (below price) to MC . Price stays above marginal cost (markup).

Optimality condition and comparison with PC

$$MR(Q_m^*) = MC(Q_m^*), \quad P_m^* > MC(Q_m^*)$$

	PC	Monopoly
Rule	$P = MC$	$MR = MC$
Quantity	Q_{pc}	$Q_m^* < Q_{pc}$
Price	P_{pc}	$P_m^* > P_{pc}$
DWL	0	> 0



Deadweight loss of monopoly

$$DWL_m = \frac{1}{2}(Q_{pc} - Q_m^*)(P_m^* - MC(Q_m^*))$$

Efficient trades *not realized* between Q_m^* and Q_{pc} .

Essential game-theory tools

Premise. In oligopoly there are *few* firms: each must anticipate the choices of the others. Game theory formalizes these strategic interactions.

Dominant strategy

s_i^* is strictly dominant for i if:

$$u_i(s_i^*, s_{-i}) > u_i(s_i, s_{-i}) \quad \forall s_i \neq s_i^*, \forall s_{-i}$$

It is the optimal choice *regardless* of what the others do.

Nash equilibrium

$(s_1^*, \dots, s_n^*)$ is a NE if no player has an incentive to deviate *unilaterally*:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall s_i, \forall i$$

Prisoner's dilemma

Dominant strategy = “confess/compete” for both; the NE is suboptimal relative to (silent, silent). Translated to oligopoly: each firm has an incentive to deviate from collusion.

Three key notions

Best response: the optimal choice given what the others do. **Nash:** mutual BRs. **Backward induction:** in sequential games, solve backward from the second stage.

Bertrand, Cournot, Stackelberg: a unified scheme

Reference market: $P = 30 - Q$, $MC_1 = MC_2 = 6$, unlimited capacity.

Bertrand (prices, simultaneous)

Firms set P_i . *Undercutting* down to MC :

$$P_1^* = P_2^* = MC = 6, \quad \pi_i = 0$$

Two firms are enough to replicate PC (*Bertrand paradox*).

Cournot (quantities, simultaneous)

Best response: $q_1 = 12 - q_2/2$, $q_2 = 12 - q_1/2$.
Symmetric equilibrium:

$$q_i^C = 8, \quad Q^C = 16, \quad P^C = 14, \quad \pi_i^C = 64$$

Generalization: $q_i^C = \frac{a-c}{(n+1)b}$.

Stackelberg (quantities, sequential)

The leader (1) moves first, the follower (2) reacts. By *backward induction*:

$$q_1^S = 12, \quad q_2^S = 6, \quad Q^S = 18, \quad P^S = 12$$

$$\pi_1^S = 72, \quad \pi_2^S = 36$$

First-mover advantage, only if the commitment is credible.

Welfare ranking

Total π and prices: Monop. ($\pi = 144$, $P = 18$) > Cournot ($P = 14$) > Stack. ($P = 12$) (note: leader profit = collusion) > Bertrand/PC ($P = 6$). More competition $\Rightarrow$ more welfare.

Collusion in repeated games

Premise. The incentive to deviate from collusion vanishes if the game is repeated *enough*: deviating pays today but jeopardizes future profits (*grim trigger*).

Sustainability condition (repeated game, discount factor δ)

Collusion is an equilibrium if

$$\delta \geq \delta^* = \frac{\pi^{\text{dev}} - \pi^{\text{coll}}}{\pi^{\text{dev}} - \pi^{\text{punish}}}$$

π^{punish} is the punishment profit: 0 in Bertrand, π^C in Cournot.

Bertrand, $n = 2$

$$\pi^{\text{coll}} = 72, \pi^{\text{dev}} = 144, \pi^{\text{punish}} = 0 \Rightarrow \delta^* = 1/2.$$

Cournot, $n = 2$

$$\pi^{\text{coll}} = 72, \pi^{\text{dev}} = 81, \\ \pi^{\text{punish}} = 64 \Rightarrow \delta^* = 9/17 \approx 0.53.$$

Effect of n : as the number of firms increases, δ^* rises $\rightarrow$ collusion is harder to sustain. This is one of the theoretical reasons why antitrust policy aims to keep the number of competitors high.

Key formulas—part 1

Concept	Formula	Condition / note
Price-taker optimum	$P = MC(Q^*)$	$MC'(Q^*) > 0$
Shutdown (SR)	$P \geq AVC_{\min}$	FC irrelevant to the decision
Indiv. supply (SR)	$Q^S(P)$: portion of MC above $AVC_{\min}$	0 otherwise
Market supply	$Q_{mkt}^S(P) = \sum_i q_i^S(P)$	horizontal sum
Supply elasticity	$\varepsilon_S = (dQ^S/dP) \cdot (P/Q^S)$	always ≥ 0
PC equilibrium	$Q^D(P^*) = Q^S(P^*)$	first welfare theorem
Consumer surplus	$CS = \frac{1}{2} Q^*(P_{\max}^D - P^*)$	triangle (linear curves)
LR, free entry	$P_{LR}^* = AC_{\min}, \pi = 0$	$n^* = Q^D/q^*$
DWL (triangle)	$\frac{1}{2}(Q^* - \tilde{Q})(P_D(\tilde{Q}) - P_S(\tilde{Q}))$	grows $\propto t^2$ with a tax
Tax incidence	cons. share = $\varepsilon_S/(\varepsilon_S - \varepsilon_D)$	the less elastic side pays more
Tax equivalence	same incidence whoever remits	accounting distinction

Key formulas—part 2

Concept	Formula	Condition / note
Monopoly optimum	$MR(Q_m^*) = MC(Q_m^*), P_m^* > MC$	$MR < P$ for $Q > 0$
<i>MR with linear D</i>	$MR = a - 2bQ$ (if $P = a - bQ$)	same intercept, twice the slope
<i>MR and elasticity</i>	$MR = P(1 - 1/ \varepsilon_D)$	the monopolist avoids the inelastic portion
Lerner index	$L = (P - MC)/P = 1/ \varepsilon_D $	at the monopolist's optimum
Monopoly DWL	$\frac{1}{2}(Q_{pc} - Q_m^*)(P_m^* - MC^*)$	efficient trades not realized
Bertrand	$P_1^* = P_2^* = MC, \pi = 0$	homogeneous good, unlim. capacity
Cournot (n firms, symm.)	$q_i^C = (a - c)/((n + 1)b)$	$P^C = (a + nc)/(n + 1)$
Stackelberg (leader)	$q_1^S = (a - c)/(2b), q_2^S = q_1^S/2$	first-mover advantage
Collusion sustainability	$\delta \geq (\pi^{dev} - \pi^{coll})/(\pi^{dev} - \pi^{punish})$	grim trigger

Graphs you must be able to draw: firm, market, policies

Firm and market

- Optimum of the PC firm: $P = MC$, profit as the area $(P - AC) \cdot Q^*$
- The four zones AVC , AC , MC (shutdown, break-even)
- Individual supply curve and horizontal aggregation
- Equilibrium (P^*, Q^*) with excesses (below/above equilibrium)
- Consumer and producer surplus (triangles)
- LR adjustment: S shifts, P returns to $AC_{\min}$

Public policies

- Price ceiling/floor: binding constraint, short side, excesses
- Specific tax: vertical shift of S , wedge t , areas (CS, PS, T , DWL)
- Ad valorem tax: rotation of S
- Subsidy: downward shift of S , cost to the government
- Tariff vs quota: who captures the trapezoid (government vs license holders)

Graphs you must be able to draw: monopoly and oligopoly

Monopoly

- Linear demand and MR with twice the slope, same intercept as D
- Optimum $MR = MC$, price read vertically off the D curve
- CS, profit, and DWL of monopoly (triangles + rectangle)
- Comparison with PC: $Q_m^* < Q_{pc}$, $P_m^* > P_{pc}$

Oligopoly

- Payoff matrix and best-response highlighting (prisoner's dilemma)
- Cournot reaction functions, intersection E^C
- The leader's move to E^S along BR_2
- Notable points in the (q_1, q_2) plane: $(q^M, 0)$, $(0, q^M)$, collusion $(q^M/2, q^M/2)$, E^C , E^S

Multiple-choice questions

1. A price-taking firm in the short run decides to produce if and only if:

- A. $P \geq AC_{\min}$
- B. $P \geq AVC_{\min}$
- C. $P \geq MC(Q^*)$
- D. $P \geq AFC(Q^*)$

Correct answer: B. Fixed costs are *sunk* in the SR and do not affect the decision to produce. In the LR the threshold becomes $P \geq AC_{\min}$.

2. A firm has $TC(q) = q^2 + 10q + 25$ and faces the price $P = 20$. The optimal quantity q^* and profit are, respectively:

- A. $q^* = 5, \pi = 0$
- B. $q^* = 5, \pi = 50$
- C. $q^* = 10, \pi = 25$
- D. $q^* = 4, \pi = 10$

Correct answer: A. $MC = 2q + 10$; $P = MC \Rightarrow q^* = 5$. $\pi = 100 - (25 + 50 + 25) = 0$: the firm is exactly at the break-even point.

Multiple-choice questions

3. The supply of a single firm is $q^s = 2P - 6$ for $P \geq 3$. If 15 identical firms operate in the market, the total quantity supplied at price $P = 10$ is:

- A. 14
- B. 60
- C. 140
- D. 210

Correct answer: D. $q^s(10) = 14$; $Q_{\text{mkt}}^s(10) = 15 \cdot 14 = 210$.

4. Given $Q^D = 60 - 2P$ and $Q^S = 4P$, consumer surplus at the competitive equilibrium is:

- A. 100
- B. 200
- C. 400
- D. 600

Correct answer: C. $60 - 2P = 4P \Rightarrow P^* = 10$, $Q^* = 40$. Inverse demand: $P_D = 30 - Q/2$, $P_{\text{max}}^D = 30$.
 $CS = \frac{1}{2} \cdot 40 \cdot (30 - 10) = 400$.

Multiple-choice questions

5. In a competitive market with free entry and exit in the long run, at the LR equilibrium:

- A. Every active firm earns positive economic profit
- B. The price equals the minimum of average total cost AC
- C. The price equals the minimum of average variable cost AVC
- D. The number of firms is given exogenously

Correct answer: B. Free entry drives P to $AC_{\min}$, with $\pi = 0$ for every active firm. n^* is endogenous and determined by $Q^D(P_{LR}^*)/q^*$.

6. A *price ceiling* (administered maximum price) is actually binding if:

- A. It is above the equilibrium price P^*
- B. It is exactly equal to the equilibrium price P^*
- C. It is below the equilibrium price P^*
- D. It is always binding, regardless of P^*

Correct answer: C. If $\bar{P} \geq P^*$ the market operates freely at P^* and the ceiling does not bite. Only $\bar{P} < P^*$ generates excess demand, rationing, and DWL.

Multiple-choice questions

7. Other things equal, a specific tax falls *mainly* on consumers if:

- A. Both demand and supply are elastic
- B. Demand is more inelastic than supply
- C. Supply is more inelastic than demand
- D. The tax is formally remitted by the seller

Correct answer: B. Cons. share = $\varepsilon_S / (\varepsilon_S - \varepsilon_D)$: it rises if $|\varepsilon_D|$ falls (more inelastic demand) or if ε_S rises. The equivalence principle rules out D.

8. For a monopolist facing downward-sloping demand, marginal revenue is strictly below price because:

- A. The monopolist always produces on the inelastic portion of demand
- B. To sell one more unit it must lower the price on *all* units
- C. Marginal cost is decreasing
- D. The demand curve is increasing in price

Correct answer: B. $MR = P + Q \cdot P'(Q) < P$ with $P'(Q) < 0$. A is inconsistent with the theory: the monopolist never produces on the inelastic portion (that would mean $MR < 0$).

Multiple-choice questions

9. A firm's Lerner index is $L = 0.4$. At the monopolist's optimum, the elasticity of the demand it faces is:

- A. -0.4
- B. -1
- C. -2.5
- D. -4

Correct answer: C. $L = 1/|\varepsilon_D| \Rightarrow |\varepsilon_D| = 1/0.4 = 2.5$, hence $\varepsilon_D = -2.5$.

10. Compared with single-price monopoly, *first-degree* (perfect) price discrimination:

- A. Generates the largest possible deadweight loss
- B. Leaves consumer surplus unchanged
- C. Captures all the surplus for the monopolist and eliminates the deadweight loss
- D. Requires consumers to self-select from a menu of tariffs

Correct answer: C. With first degree, π = the entire area between D and MC (from 0 to Q_{pc}); $CS = 0$ and $DWL = 0$. Self-selection is second degree.

Multiple-choice questions

11. In the Bertrand duopoly with a homogeneous good, common marginal cost MC , unlimited capacity, and one-shot interaction, the equilibrium is:

- A. $P_1 = P_2 = P^M$ (monopoly price), profit $\pi^M/2$ each
- B. $P_1 = P_2 = MC$, zero profits for both
- C. One firm sets MC , the other the monopoly price
- D. No equilibrium in pure strategies

Correct answer: B. For any $P > MC$ each firm has an incentive to undercut; the only rest point is $P = MC$. Two firms are enough to replicate PC (*Bertrand paradox*).

12. In Akerlof's "lemons" model, the effect of adverse selection is that:

- A. Only high-quality cars remain on the market
- B. Both good and bad cars are traded at their true values
- C. Low-quality cars tend to prevail, possibly leading to the collapse of the market for good cars
- D. The price perfectly reflects hidden quality

Correct answer: C. Sellers of good cars withdraw if the average price offered does not cover their reservation value: the composition of the pool worsens, the price falls further, possibly until efficient trade collapses.

Exercise 1

A price-taking firm operates in a competitive market with short-run total cost function:

$$TC(q) = q^2 + 4q + 9$$

- (a) Derive the marginal cost curve $MC(q)$, average variable cost $AVC(q)$, and average total cost $AC(q)$. Find the **minimum average cost** $AC_{\min}$ and the quantity q^* at which it is reached.
- (b) Write the **individual short-run supply curve** $q^s(P)$ and compute the optimal quantity and profit for $P = 14$.
- (c) If $N = 20$ identical firms operate in the market, write the **market supply curve** $Q^S(P)$.
- (d) Suppose market demand is $Q^D = 90 - 3P$ and that there is free entry and exit. Determine the **long-run equilibrium**: price P_{LR}^* , market quantity Q_{LR}^* , and number of firms n^* .

Exercise 1—Solution (a)

(a) Cost functions and minimum of AC

$$MC(q) = \frac{dTC}{dq} = 2q + 4, \quad VC(q) = q^2 + 4q, \quad AVC(q) = q + 4, \quad AC(q) = q + 4 + \frac{9}{q}$$

Minimum of AC: $dAC/dq = 1 - 9/q^2 = 0 \Rightarrow q^2 = 9 \Rightarrow \boxed{q^* = 3}$.

$$AC(3) = 3 + 4 + \frac{9}{3} = 3 + 4 + 3 = \boxed{10}$$

Check $MC = AC$ at q^* : $MC(3) = 2 \cdot 3 + 4 = 10 = AC(3) \checkmark$.

Note on AVC

$AVC(q) = q + 4$ is monotonically increasing: $AVC_{\min} = \lim_{q \rightarrow 0^+} AVC(q) = 4$. The shutdown threshold is therefore $P = 4$.

Exercise 1—Solution (b)

(b) Individual supply curve

From the condition $P = MC(q)$:

$$P = 2q + 4 \implies q^s(P) = \frac{P - 4}{2} \quad \text{for } P \geq 4$$

$$q^s(P) = 0 \quad \text{for } P < 4$$

Case $P = 14$

$$q^s(14) = \frac{14 - 4}{2} = \boxed{5}$$

Profit at $P = 14$

$$\pi = P \cdot q - TC(q) = 14 \cdot 5 - (25 + 20 + 9) = 70 - 54 = \boxed{16}$$

Economic interpretation

$P = 14 > AC(5) = 5 + 4 + 9/5 = 10.8$: the firm is in the positive-profit region (Zone 4). In the LR, $\pi > 0$ would attract new firms: the price is not sustainable in steady state.

Exercise 1—Solution (c)

(c) Market supply curve with $N = 20$ identical firms

$$Q^S(P) = N \cdot q^s(P) = 20 \cdot \frac{P - 4}{2} = 10(P - 4) = \boxed{10P - 40} \quad \text{for } P \geq 4$$

Inverse form: $P = Q^S/10 + 4$.

Interpretation

The price threshold at which the market starts supplying is $P = 4$ (the same as for the single firm). For each additional euro of price, market quantity rises by 10 units (sum over 20 firms, each contributing 1/2).

Exercise 1—Solution (d)

(d) Long-run equilibrium with free entry

With free entry, the LR price is the minimum of average cost:

$$P_{LR}^* = AC_{\min} = 10$$

Each active firm produces at efficient scale $q^* = 3$ with $\pi = 0$. **Total market quantity from demand**

$$Q_{LR}^* = Q^D(10) = 90 - 3 \cdot 10 = 60$$

Number of firms in the LR

$$n^* = \frac{Q_{LR}^*}{q^*} = \frac{60}{3} = 20$$

Checking zero profit

$$\pi = P_{LR}^* \cdot q^* - TC(q^*) = 10 \cdot 3 - (9 + 12 + 9) = 30 - 30 = 0 \checkmark.$$

Exercise 2

In a competitive market, market demand and supply are:

$$Q^D(P) = 100 - 2P, \quad Q^S(P) = 3P$$

- (a) Determine the **equilibrium price and quantity** (P^*, Q^*) .
- (b) Compute **consumer surplus**, **producer surplus**, and **total welfare**.
- (c) The government introduces a **specific tax** $t = 5$ € levied on the seller. Determine the new price paid by the consumer P_d^* , the price received by the producer P_s^* , the quantity traded Q_t^* , the **tax revenue** T , and the **deadweight loss** DWL.
- (d) Compute the **incidence shares** on consumers and producers and verify the result using the **elasticities** of D and S at the initial equilibrium.

Exercise 2—Solution (a) and (b)

(a) Competitive equilibrium

Set $Q^D = Q^S$:

$$100 - 2P = 3P \implies 100 = 5P \implies P^* = 20$$

$$Q^* = 3 \cdot 20 = 60$$

(b) Surplus and welfare

Inverse demand: $P_D = 50 - Q/2$, so $P_{\max}^D = 50$. Inverse supply: $P_S = Q/3$, so $P_{\min}^S = 0$.

$$CS = \frac{1}{2} \cdot Q^* \cdot (P_{\max}^D - P^*) = \frac{1}{2} \cdot 60 \cdot (50 - 20) = \frac{1}{2} \cdot 60 \cdot 30 = 900$$

$$PS = \frac{1}{2} \cdot Q^* \cdot P^* = \frac{1}{2} \cdot 60 \cdot 20 = 600$$

$$TS = CS + PS = 900 + 600 = 1500$$

Exercise 2—Solution (c)

(c) Specific tax $t = 5$

The new supply as a function of P_d : $Q_t^S = 3(P_d - 5) = 3P_d - 15$. Setting it equal to Q^D :

$$100 - 2P_d = 3P_d - 15 \implies 115 = 5P_d \implies P_d^* = 23$$

$$P_s^* = P_d^* - t = 23 - 5 = 18, \quad Q_t^* = 100 - 2 \cdot 23 = 54$$

Revenue and DWL

$$T = t \cdot Q_t^* = 5 \cdot 54 = 270$$

$$DWL = \frac{1}{2}(Q^* - Q_t^*) \cdot t = \frac{1}{2} \cdot 6 \cdot 5 = 15$$

Checking the welfare balance

$$CS' = \frac{1}{2} \cdot 54 \cdot (50 - 23) = 729; \quad PS' = \frac{1}{2} \cdot 54 \cdot 18 = 486; \quad \text{sum:}$$

$$CS' + PS' + T + DWL = 729 + 486 + 270 + 15 = 1,500 = TS \checkmark$$

Exercise 2—Solution (d)

(d) Incidence shares

$$\begin{aligned}\Delta P_d &= 23 - 20 = 3, & |\Delta P_s| &= 20 - 18 = 2 \\ \text{Consumers' share} &= \frac{\Delta P_d}{t} = \frac{3}{5} = 60\%, & \text{Producers' share} &= \frac{|\Delta P_s|}{t} = \frac{2}{5} = 40\%\end{aligned}$$

Check with the elasticities at the initial equilibrium

$$\begin{aligned}\varepsilon_D &= \frac{dQ^D}{dP} \cdot \frac{P^*}{Q^*} = -2 \cdot \frac{20}{60} = -\frac{2}{3}, & \varepsilon_S &= 3 \cdot \frac{20}{60} = 1 \\ \text{Cons. share} &= \frac{\varepsilon_S}{\varepsilon_S - \varepsilon_D} = \frac{1}{1 - (-2/3)} = \frac{1}{5/3} = \frac{3}{5} = 60\% \checkmark\end{aligned}$$

Interpretation

Demand is *more inelastic* than supply ($|\varepsilon_D| = 2/3 < 1 = \varepsilon_S$): consumers bear the larger share of the tax. The same result would obtain if the tax were formally levied on the consumer (equivalence principle).

Exercise 3

A market is characterized by the inverse demand

$$P = 80 - Q$$

and by constant marginal costs equal for all firms: $MC = 20$ (no fixed costs).

- (a) Suppose the market is served by a **monopolist**. Compute Q_m^* , P_m^* , profit π_m , and the deadweight loss DWL relative to the perfect competition benchmark.
- (b) Now suppose a **Cournot duopoly** with two identical firms. Derive the reaction functions, the Cournot equilibrium (q_1^C, q_2^C) , and individual profits.
- (c) Finally, suppose a **Stackelberg duopoly** with firm 1 as leader. Compute q_1^S , q_2^S , price, total quantity, and profits.
- (d) Build a **summary table** with Q , P , and total π for the three regimes (monopoly, Cournot, Stackelberg) and comment.

Exercise 3—Solution (a): monopoly

(a) The monopolist's optimum

$TR = P \cdot Q = (80 - Q)Q$, so $MR = 80 - 2Q$. Condition $MR = MC$:

$$80 - 2Q = 20 \implies \boxed{Q_m^* = 30}, \quad P_m^* = 80 - 30 = \boxed{50}$$

$$\pi_m = (P_m^* - MC) \cdot Q_m^* = (50 - 20) \cdot 30 = \boxed{900}$$

PC benchmark and DWL

Under perfect competition: $P = MC \Rightarrow 80 - Q = 20 \Rightarrow Q_{pc} = 60, P_{pc} = 20$.

$$DWL_m = \frac{1}{2}(Q_{pc} - Q_m^*)(P_m^* - MC) = \frac{1}{2} \cdot 30 \cdot 30 = \boxed{450}$$

Lerner index at the optimum

$L = (P_m^* - MC)/P_m^* = 30/50 = 0.6$; consistent with $|\varepsilon_D| = 1 \cdot 50/30 = 5/3$ and $1/|\varepsilon_D| = 3/5 = 0.6 \checkmark$.

Exercise 3—Solution (b): Cournot

(b) Profits and reaction functions

$$\pi_1(q_1, q_2) = (80 - q_1 - q_2 - 20) q_1 = (60 - q_1 - q_2) q_1$$

FOC for firm 1: $\partial \pi_1 / \partial q_1 = 60 - 2q_1 - q_2 = 0$:

$$q_1 = 30 - \frac{q_2}{2}$$

By symmetry, $q_2 = 30 - q_1/2$. **Symmetric equilibrium**

Substituting:

$$q_1 = 30 - \frac{1}{2}(30 - q_1/2) = 15 + q_1/4 \implies \frac{3}{4}q_1 = 15 \implies q_1^C = q_2^C = 20$$

$$Q^C = 40, \quad P^C = 80 - 40 = 40$$

$$\pi_1^C = \pi_2^C = (40 - 20) \cdot 20 = 400$$

Exercise 3—Solution (c): Stackelberg

(c) Backward induction, leader = firm 1

Stage 2 (follower): firm 2's Cournot BR remains valid:

$$q_2^*(q_1) = 30 - \frac{q_1}{2}$$

Stage 1 (leader): firm 1 substitutes the follower's BR into its own profit:

$$\pi_1 = (60 - q_1 - q_2) q_1 = \left(60 - q_1 - 30 + \frac{q_1}{2}\right) q_1 = \left(30 - \frac{q_1}{2}\right) q_1$$

FOC: $d\pi_1/dq_1 = 30 - q_1 = 0 \Rightarrow \boxed{q_1^S = 30}$. **Follower's reaction and market outcome**

$$q_2^S = 30 - \frac{30}{2} = \boxed{15}, \quad Q^S = 45, \quad P^S = 80 - 45 = \boxed{35}$$

$$\pi_1^S = (35 - 20) \cdot 30 = \boxed{450}, \quad \pi_2^S = (35 - 20) \cdot 15 = \boxed{225}$$

Exercise 3—Solution (d): summary table

(d) Summary table of the three regimes (plus PC benchmark)

Regime	q_1	q_2	Q	P	Total π
Monopoly (cartel)	—	—	30	50	900
Stackelberg (1 leader)	30	15	45	35	675
Cournot	20	20	40	40	800
Perfect competition	—	—	60	20	0

Effect of competition: moving from cartel $\rightarrow$ Cournot $\rightarrow$ Stackelberg $\rightarrow$ PC, total quantity rises and the price falls. More competition $\Rightarrow$ greater consumer welfare, lower aggregate industry profit.

Exercise 3—Solution (d): comment

Key takeaways from the comparison:

- The Stackelberg leader *earns exactly half of the cartel profit* (450 vs 900), but more than under Cournot (450 vs 400);
- The follower is the big loser: 225 vs 400 under Cournot;
- The leader prefers Stackelberg to Cournot; the follower prefers Cournot to Stackelberg.

First-mover advantage

The asymmetry in Stackelberg exists *only* if the leader's commitment is observable and credible (installed capacity, multi-year contracts, irreversible investments). Without commitment, the only consistent equilibrium is Cournot.

Bonus

Rent-seeking and rent dissipation.

Consider a monopolist that obtains its market position thanks to an *exclusive license* granted by the government. Several contenders have spent substantial resources to obtain it (lobbying, advertising campaigns, litigation, consulting).

- (a) Define the concept of **rent-seeking** and explain why it can represent a welfare loss *in addition* to the classic deadweight loss of monopoly (Harberger triangle). *Five lines maximum.*
- (b) According to Tullock's argument (1967), in a fully competitive and symmetric rent-seeking setting, what is the limiting value of the resources dissipated in total by the contenders trying to obtain the rent? What does this result imply for the overall social cost of monopoly? *Five lines maximum.*

Hint

No calculations are required. The answer is purely conceptual: what the phenomenon is, why it adds a loss, where it stands relative to the DWL triangle.

Bonus

(a) Definition and additional loss

Rent-seeking is the set of *unproductive* activities aimed at obtaining, maintaining, or extending an *economic rent* generated by artificial restrictions on competition (licenses, patents, tariffs, regulation). The resources spent on these activities (lobbying, litigation, corruption, consulting) are *diverted from productive uses*: their unproductive use is a welfare loss **in addition** to the allocative DWL (Harberger triangle). In other words, the monopoly rent is not a pure transfer from consumers to the producer: it is partly *burned* in the attempt to capture it.

(b) Tullock's limit

In Tullock's model, if many contenders compete symmetrically for the rent with constant marginal returns, in equilibrium total rent-seeking expenditure **equals the value of the rent itself**: the rent is *fully dissipated*. The true social cost of monopoly is therefore not just the Harberger triangle, but also the *entire monopoly profit rectangle*: the monopolist gets the rent but has already spent it to obtain it, and no one (neither consumers nor producer) benefits.

Economic interpretation

Tullock's result is an *upper bound* on dissipation: in reality dissipation is partial (because of information asymmetries, institutional constraints, rising costs). Yet the policy conclusion stands: government-created rents are not inert—they generate real costs, the larger the more open the competition for them.

Summary of common exam mistakes

Firm and supply

- Confusing shutdown ($AVC_{\min}$, SR) with break-even ($AC_{\min}$, LR)
- Identifying the supply curve with the *whole* MC instead of only its rising portion above $AVC_{\min}$
- Summing individual supply curves vertically (rather than horizontally)

Equilibrium and welfare

- Computing CS/PS with the slope instead of the intercept
- Treating DWL as a transfer rather than as value that vanishes

Taxes and monopoly

- Confusing who remits the tax with who bears its burden (equivalence principle)
- Applying $P = MC$ to the monopolist (it holds only under PC)
- Using the height $(P_m^* - P_{pc})$ instead of $(P_m^* - MC^*)$ in the monopoly DWL

Oligopoly

- Applying the monopoly condition to Cournot (ignoring the rival's BR)
- Confusing Cournot and Stackelberg: in Stackelberg, substitute BR_2 *before* maximizing π_1
- Forgetting that collusion holds in the repeated game only if $\delta \geq \delta^*$